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Batch - 4(OS)

LSTM Practice Questions

Q1

Input $x_t = 0.5$

Previous hidden state $h_{t-1} = 0.2$

Prev cell state $C_{t-1} = 0.4$

$$\sigma(2) = \frac{1}{1 + e^{-2}}$$

$$\tanh(2) = \frac{e^2 - e^{-2}}{e^2 + e^{-2}}$$

i) forget gate: $f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$

Substitute values:

$$f_t = \sigma(0.7(0.5) + 0.3(0.2) + 0.1)$$

$$= \sigma(0.510)$$

sigmoid: $\frac{1}{1 + e^{-0.5100}}$

$$\approx 0.6248$$

$$f_t = 0.6248 //$$

The forget gate retains approximately 62.48 % of the prev. cell state information.

2) input gate : $i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$

$i_t = \sigma(0.6(0.5) + 0.4(0.2) + 0.0)$

$i_t = \sigma(0.3800)$

Sigmoid : $\frac{1}{1 + e^{-0.3800}}$

$i_t = 0.5938$ //

The input gate controls how much new candidate information enters the cell state

3) Candidate State : $\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$

substitute values:

$\tilde{C}_t = \tanh(0.9(0.5) + 0.2(0.2) + 0.0)$

$\tilde{C}_t = \tanh(0.4900)$

$\tilde{C}_t = \frac{e^{0.4900} - e^{-0.4900}}{e^{0.4900} + e^{-0.4900}}$

$\boxed{\tilde{C}_t = 0.4542}$

4) New cell state C_t :

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

$$C_t = (0.6248)(0.4) + (0.5938)(0.4542)$$

$$\boxed{C_t = 0.5196}$$

The new cell state contains both retained information from the prev state and new candidate info.

5) output gate : O_t

$$O_t = \sigma(W_o x_t + V_o h_{t-1} + b_o)$$

$$O_t = \sigma(0.4)$$

$$O_t = \frac{1}{1 + e^{-0.4}}$$

$$\boxed{O_t = 0.6100}$$

6) Hidden State h_t

$$h_t = O_t \tanh(C_t)$$

$$h_t = 0.6100 \tanh(0.5196)$$

$$\boxed{h_t = 0.2912}$$

Q2

Input Dim D = 10

Hidden units H = 32

a) One bias vector per gate

for each gate, the trainable parameters are:

1. Input weight : DxH
2. Recurrent weight : HxH
3. Bias : H

for 1 gate : $P_{gate} = DH + H^2 + H$

$P_{gate} = (10)(32) + (32)(32) + 32$

$P_{gate} = 1376$

for 4 gates : $P_{total} = 4 P_{gate}$

$= 4(1376)$

$P_{total} = 5504$

b) Two bias vector per gate

$P_{gate} = DH + H^2 + 2H$

$P_{gate} = 320 + 1024 + 64$

$= 1408$

for 4 gates : $P_{total} = 4(1408)$

$P_{total} = 5632$

Q3 The gradient scaling factor is approximated by:

Scaling factor = f^k

f = Constant forget gate value

k: no of time step.

a) $f = 0.9$

After 10 steps:

Scaling factor = 0.9^{10}

= 0.34867844

Rounded to 4 decimal places:

0.3487

The gradient retains appox 34.87% of its original magnitude

After 50 steps:

S.F = 0.9^{50}

≈ 0.00515378

= 0.0052

The gradient retains approx 0.52% of its original magnitude.

b) $k = 50$

Case 1: $f = 0.5$

SF = 0.5^{50}

= 8.8818×10^{-16}

= 0.0000

The result is zero, although the actual value is extremely small and not exactly zero.

Ex 2:- $f = 0.99$

$SF = 0.99^{50}$

≈ 0.605

The gradient retains approx 60.50% of its original magnitude.

c) Interpretation :-

When the forget gate is 0.5 or 0.9 the gradient becomes very small over 50 time steps, making it difficult for the network to learn long term dependencies.

When the forget gate is 0.99 the gradient retains above 60.50% of its magnitude. LSTM helps reduce vanishing gradient through their cell state and learned forget gate which can preserve information over long sequences; they do not eliminate the problem ~~process~~ problem in every situation.

Q4.

a) Total hourly records

one day contains 24 hours

one year : 365 × 24 = 8760 hrs.

Three years : N = 3 × 8760

N = 26,280 records

b) no. of input / output windows

Each window requires

168 input records

24 subsequent target records.

with a stride of 1 hr, the no. of valid windows is :

windows = N − lookback − horizon + 1

= 26280 − 168 − 24 + 1

= 26089

The +1 accounts for the first valid window.

c) Tensor Shapes for a batch of 64 windows

The standard input shape is

(batch size, time steps, features)

X = (64, 168, 3)

X : (64, 168, 3)

for 24 hr forecast assuming the model predicts one load value per hour

$$y: (64, 24)$$

This means:

- 64 separate windows.
- each ~~step~~ input contains 168 hourly records and 3 features per record
- each op contains 24 predicted hourly load values

d) Training window and mini-batches

The first 70% of records are used for training

Step 1:- Calculate training records.

$$N_{train} = 0.70 \times 26280$$
$$= 18,396$$

Step 2 :- Calculate training windows.

$$W_{train} = N_{train} - 168 - 24 + 1$$
$$= 18,205$$

Step 3:- Calculate mini-batches

$$\text{batches} = \left\lfloor \frac{\text{training windows}}{\text{Batch size}} \right\rfloor$$

$$= \frac{18205}{64}$$

Size: $64 \times 284 = 18176$

Remaining windows: 18205 − 18176 = 29

There are 284 full batches and one final partial batch

285 mini-batches per epoch

e) LSTM parameter count

Given: input dim $D = 3$

hidden size $H = 64$

one bias vector per gate

$$P = 4H(D + H + 1)$$

$$P = 4(64)(3 + 64 + 1)$$

$$\boxed{P = 17,408}$$ trainable parameters

Q5

(a) Min-Max Normalization

Given: min reading = 20 mm/s

max " = 120 mm/s

actual " = 95 mm/s

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$= \frac{95 - 20}{120 - 20}$$

$$\boxed{x_{norm} = 0.7500}$$

normalized reading as 0.75, lies in range 0 and 1

b) failure prediction using threshold

Given:

Predicted failure prob = 0.83

decision threshold = 0.5

Decision rule:

$\hat{y} = 1, \quad P \geq 0.5$

$0, \quad P < 0.5$

Compare

$0.83 \geq 0.5$

Prediction = 1 (failure)

The model predicts that the pump will fail within the next cycle.

c) confusion matrix and evaluation matrix.

given total cycles = 1000

actual failures = 40

predicted " = 45

correctly " = 30

TP = 30

FP = Predicted failure - TP

= 45 - 30

FP = 15

→ FN = actual failure - TP

= 40 - 30

FN = 10

→ TN = 1000 - 40 - 15

TN = 945

Confusion matrix :

actual / Predicted	Failure	No failure	total
failure	TP = 30	FN = 10	40
no failure	FP = 15	TN = 945	960
Total	45	955	1000

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{30}{30+15} = \frac{30}{45}$$

$$= 66.67\% \mid 0.6667$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\frac{30}{30+10}$$

$$\frac{30}{40}$$

$$= 0.7500 \mid 75\%$$

F1 Score = $\frac{2PR}{P+R}$

$$\frac{2(30/45)(30/40)}{\left(\frac{30}{45}\right) + \left(\frac{30}{40}\right)}$$

$F_1 = \frac{60}{85}$

$= 0.70509 \mid 70.59\%$

$\text{accuracy} = \frac{TP + TN}{N}$

$= \frac{30 + 945}{1000}$

$= \frac{975}{1000}$

$= 97.50\%$

d) The maintenance team should generally prioritize recall because missing a genuine pump failure can lead to equipment damage, costly downtime, or safety risk

Recall is 75% which means the model detects 30 out of 40 actual failures and misses 10.

Accuracy can be misleading because only 40 out of 1000 cycles are followed by a failure. A model that predicts no failure for every cycle would achieve:

$$\text{accuracy} = \frac{960}{1000} \times 100$$

$$= 96\%$$

Yet it would detect zero actual failure giving a recall of 0%.

∴ accuracy alone is not enough the team should evaluate recall alongside precision, recall, F1-score, and the operational cost of false alarm